

Results Overview

- **Designs generated: 18012**

- Designs passing all filters: 12988

- **MMCIF files of final designs are in:**

/home/maom/turbo/projects/MOR_ultra/intermediate/boltzgen/morphine/merged/final_ranked_designs/final_30_designs

- **Metrics and sequences of 30 final designs are in:**

/home/maom/turbo/projects/MOR_ultra/intermediate/boltzgen/morphine/merged/final_ranked_designs/final_designs_metrics_30.csv

- **Metrics of all designs are in:**

/home/maom/turbo/projects/MOR_ultra/intermediate/boltzgen/morphine/merged/final_ranked_designs/all_designs_metrics.csv

You can rerun filtering (very quick), using this command with changed parameters:

```
! boltzgen run input_spec.yaml --steps filtering --config filtering budget=60 alpha=0.05
```

You can also rerun filtering in a jupyter notebook if you want using `filter.ipynb``

What was run to produce this in the Filter task:

1. Filtering: each design is evaluated against mandatory thresholds.
2. Ranking: for every metric we compute its rank, then scale it by the metric's inverse-importance weight.

Designs with fewer passed filters are automatically penalised because the ranking key is the pair (num_filters_passed, metric). The overall quality score is the worst (maximum) of these scaled ranks.

The 10 best designs form the Top set.

3. Diversity: a lazy-greedy algorithm selects 30 designs that jointly maximise quality and minimise sequence similarity (sequence-identity distance). The trade-off is controlled by $\alpha = 0.001$:

- $\alpha = 0 \rightarrow 100\%$ quality focus (same as Top set)
- $\alpha = 1 \rightarrow 100\%$ diversity focus (ignores quality)
- Quality – composite of metrics such as iPTM, salt-bridges, Δ SASA, etc. Each metric has an "inverse importance" weight (see "Sorting Criteria" table). A larger weight divides the rank by a bigger number and therefore down-weights that metric.
- Diversity – 1 – sequence identity between designs.

We use $\alpha = 0.001$, meaning 100 % emphasis on quality, 0 % on diversity.

CSV Column Reference

Column	Description
id	filename to retrieve the file
design_sequence	designed amino acids (may be subset of chain)
designed_chain_sequence	full sequence of the chain containing designed residues (recommended for synthesis)
num_design	number of designed residues
secondary_rank	intermediate rank from the sorting procedure
design_ptm	predicted TM score for intra-design contacts (higher = better)
design_ip tm	predicted TM score for design-target contacts (higher = better)
design_to_target_ip tm	same as design_ip tm but for multi-chain designs
min_design_to_target_pae	minimum PAE between design & target (lower = better)
plip_saltbridge	number of salt-bridge interactions
plip_hbonds	number of hydrogen-bond interactions
plip_hydrophobic	number of hydrophobic interactions
delta_sasa_original	Δ SASA when binder present vs absent
delta_sasa_refolded	same as above but on the refolded structure

Filtering & Sorting Criteria

This section first lists threshold filters that every design must pass (Filtering Criteria table) then explains how designs are ranked (Sorting Criteria table) weighted by their inverse importance.

Filtering Criteria

feature	lower_is_better	threshold	Pass
has_x	True	0.0	18012
filter_rmsd	True	2.5	15503
filter_rmsd_design	True	2.5	15503
designfolding-filter_rmsd	True	2.5	14793
ALA_fraction	True	0.3	15827
GLY_fraction	True	0.2	17499
GLU_fraction	True	0.2	17737
LEU_fraction	True	0.3	18007
VAL_fraction	True	0.2	17902

Sorting Criteria

Metric	Inverse Importance
design_to_target_ip tm	1
design_ptm	1
neg_min_design_to_target_pae	1
plip_hbonds_refolded	2
plip_saltbridge_refolded	2
delta_sasa_refolded	2

Results Summary

Aggregate statistics

Quick numeric overview comparing (i) ALL incoming designs, (ii) the top-10 highest-quality designs, and (iii) the 30 quality+diversity designs produced by the lazy-greedy selection.

Metric	Mean	Mean top 10	Mean top 30 diverse
Num designs	18012	-	-
num_design	108.512	107.600	107.633
filter_rmsd	1.841	0.721	0.813
designfolding-filter_rmsd	2.123	1.023	0.982
design_iptm	0.587	0.806	0.775
design_to_target_iptm	0.587	0.806	0.775
min_design_to_target_pae	4.663	1.849	2.093
delta_sasa_refolded	296.935	404.563	379.966
plip_saltbridge_refolded	0.000	0.000	0.000
plip_hbonds_refolded	1.138	3.000	2.567

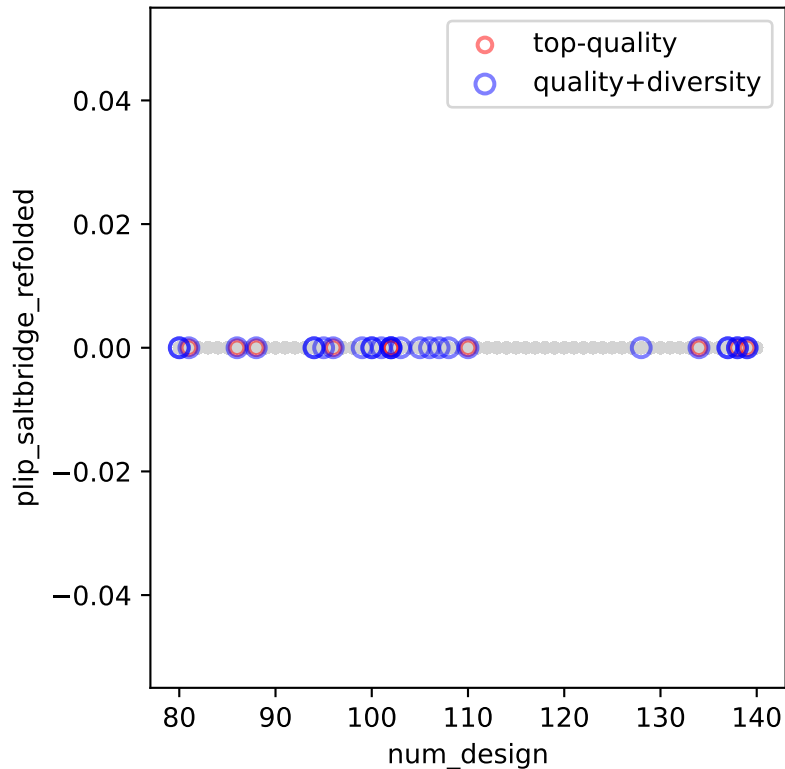
Scatter Plots

Metric relationships

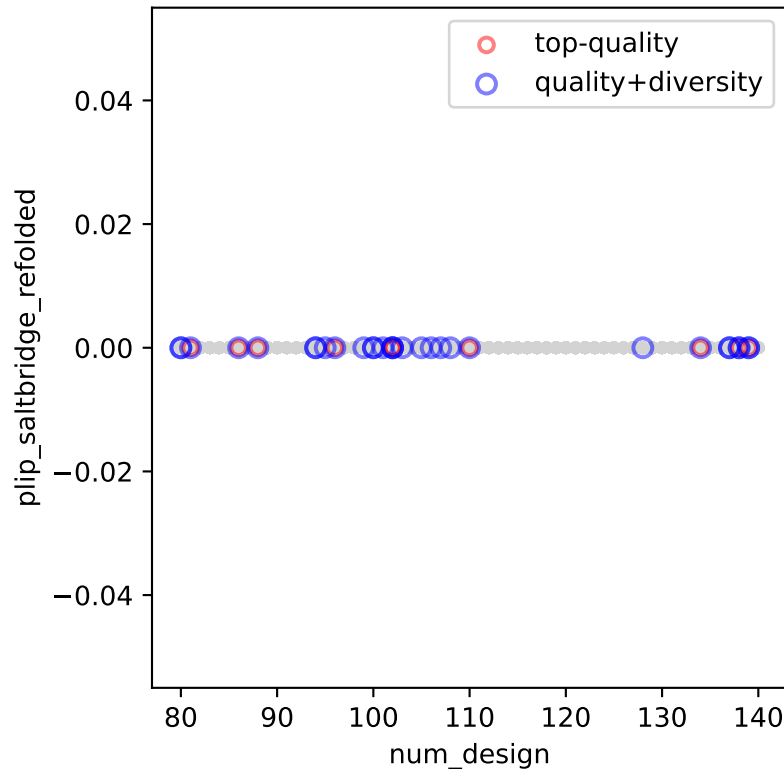
Each scatter page contains two panels:

- Left – all designs (grey) with overlays of Top (red) and Diverse (blue).
- Right – same but limited to designs passing the RMSD filter.

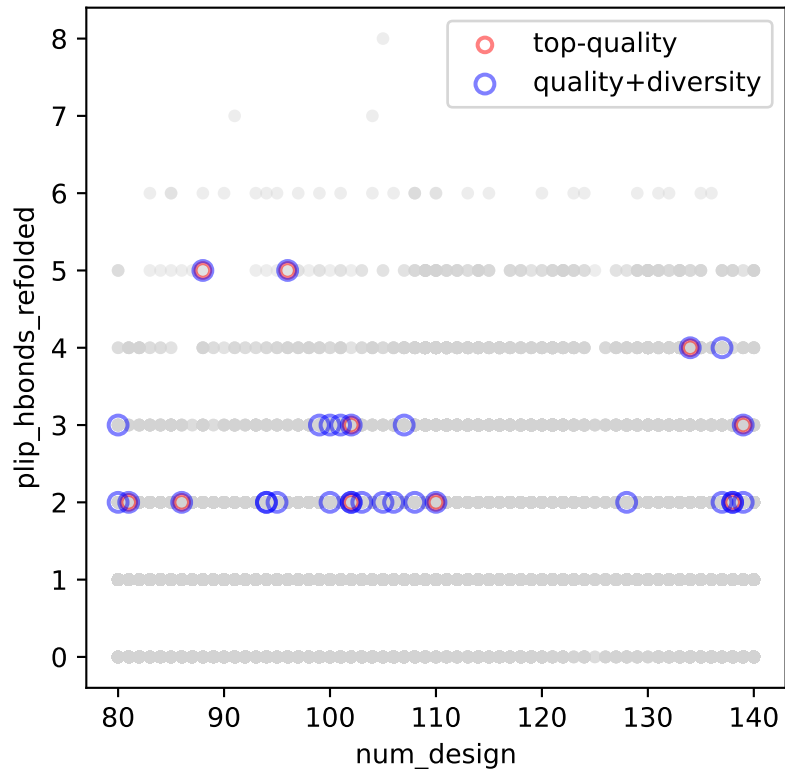
All samples



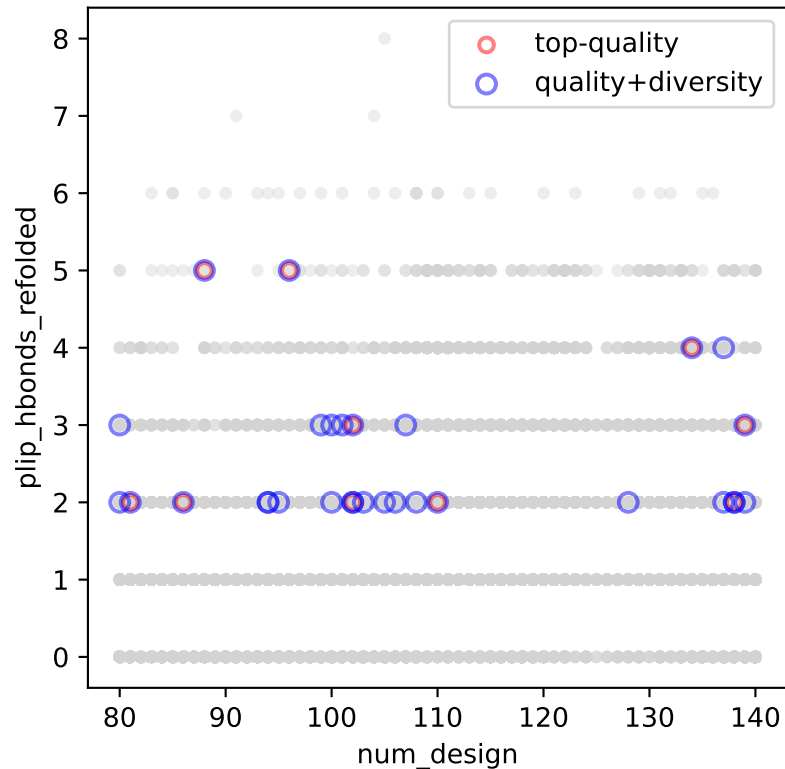
(Designs passing RMSD threshold)



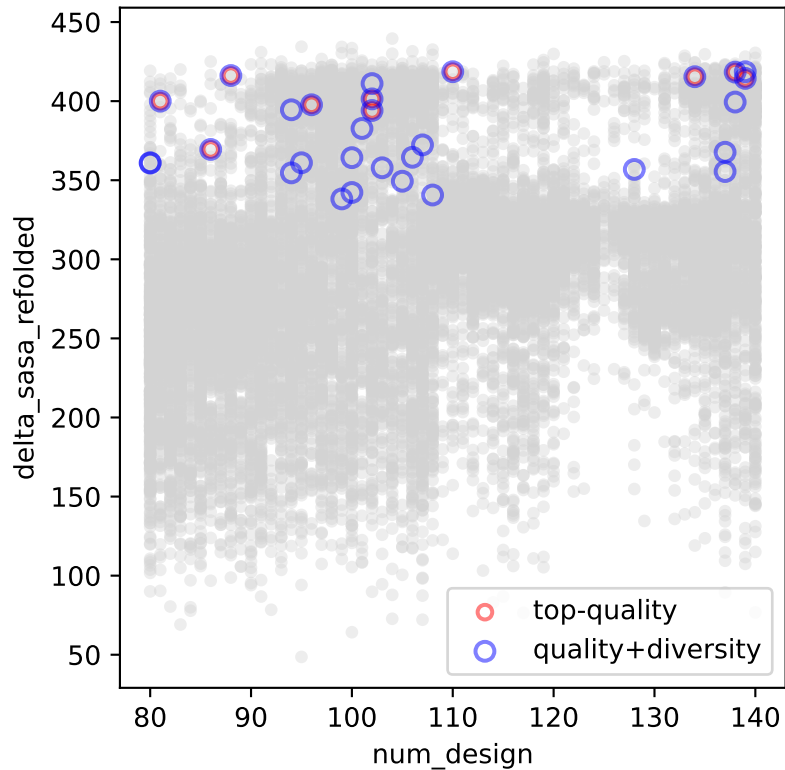
All samples



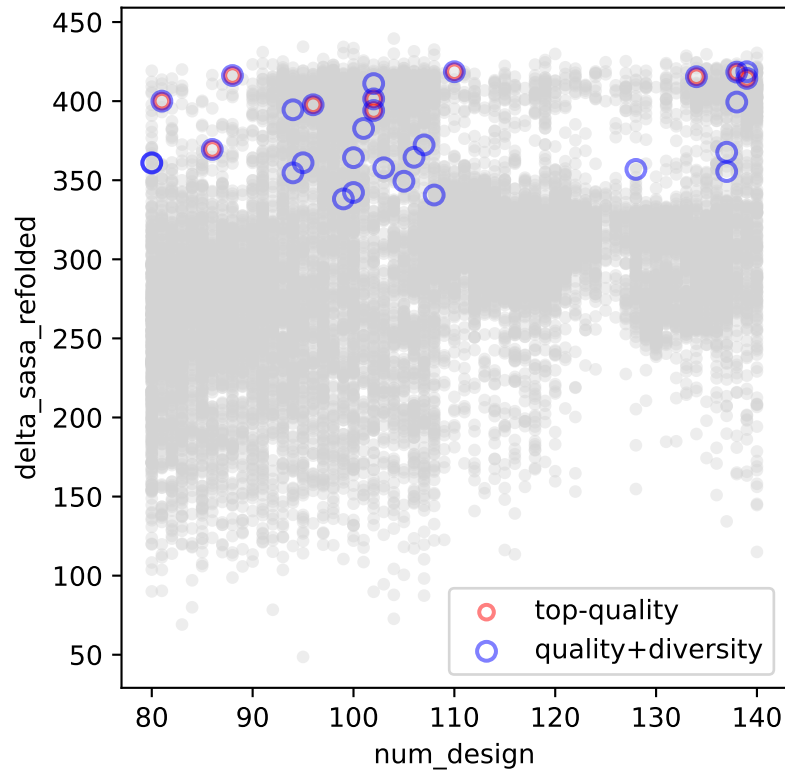
(Designs passing RMSD threshold)



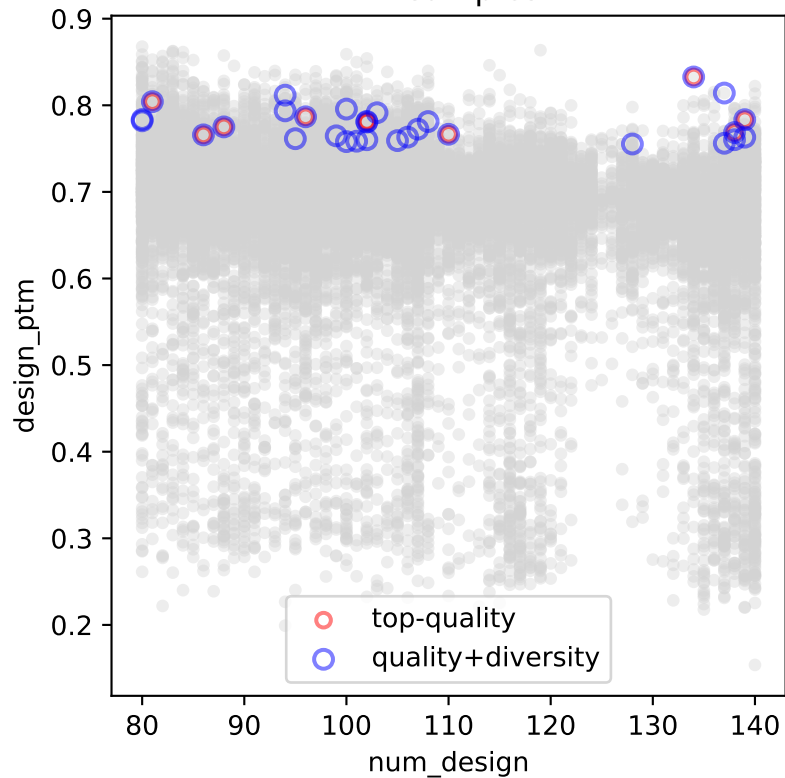
All samples



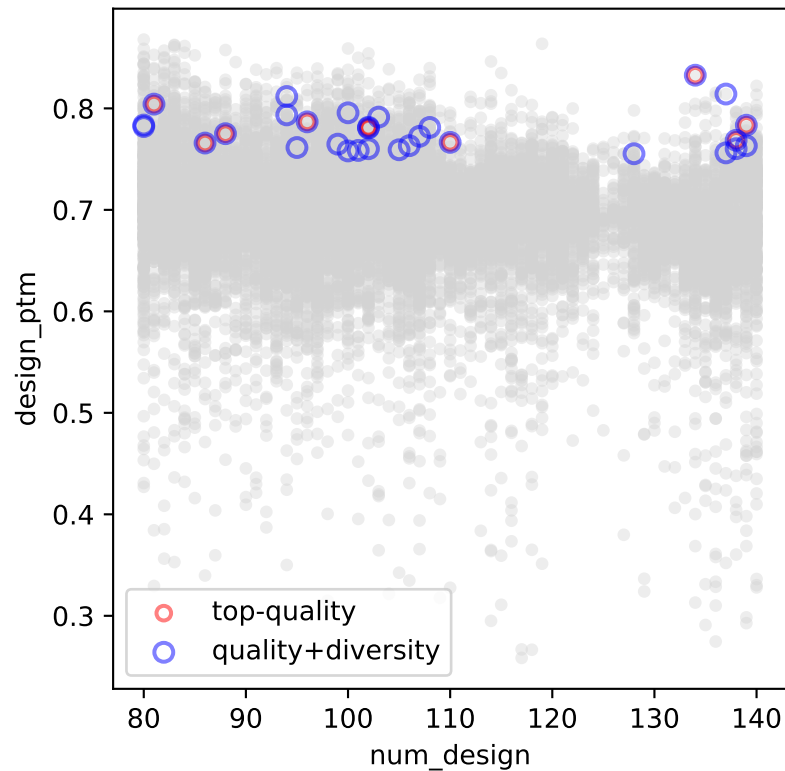
(Designs passing RMSD threshold)



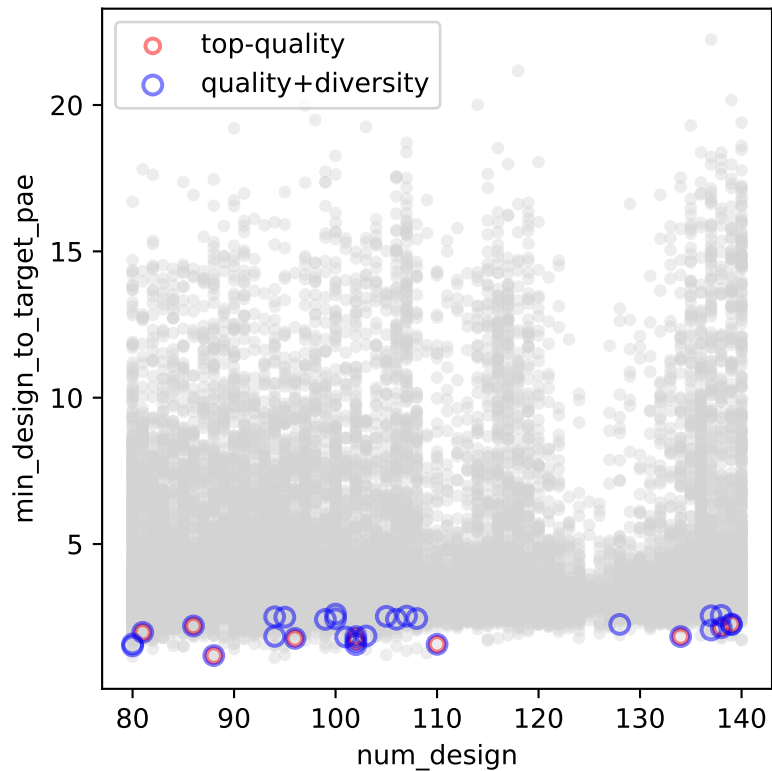
All samples



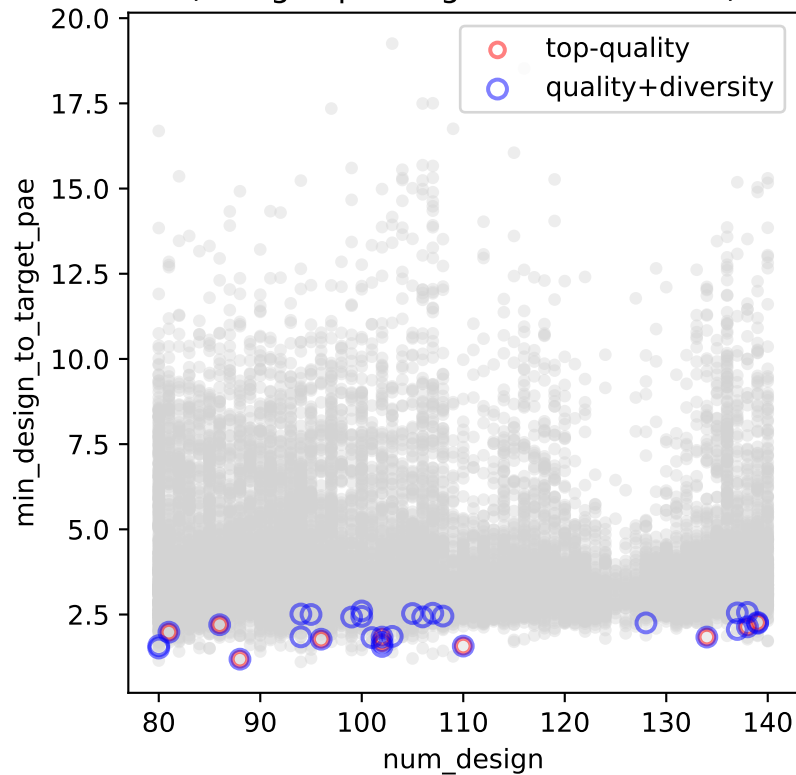
(Designs passing RMSD threshold)



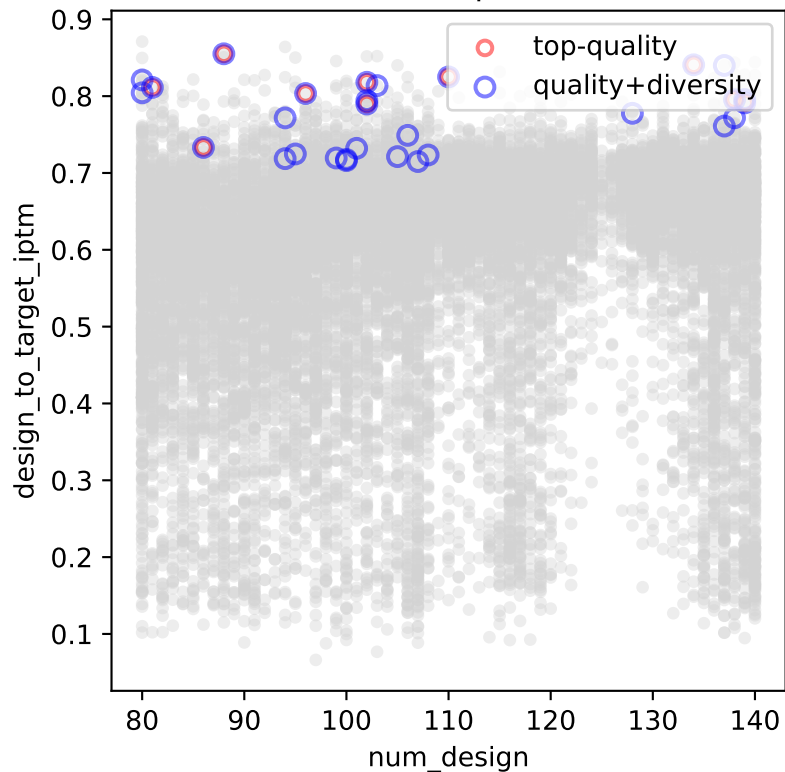
All samples



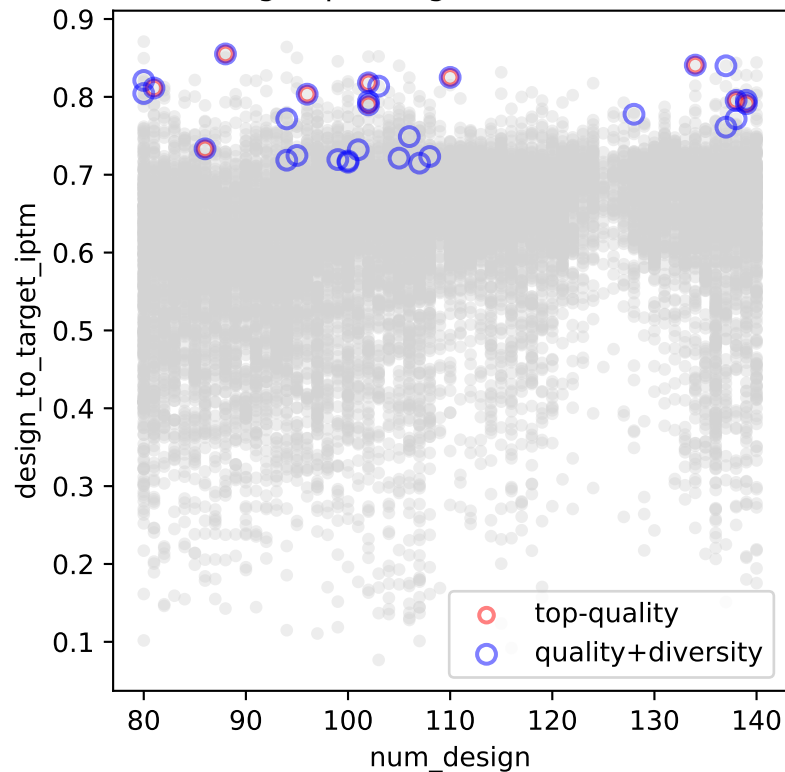
(Designs passing RMSD threshold)



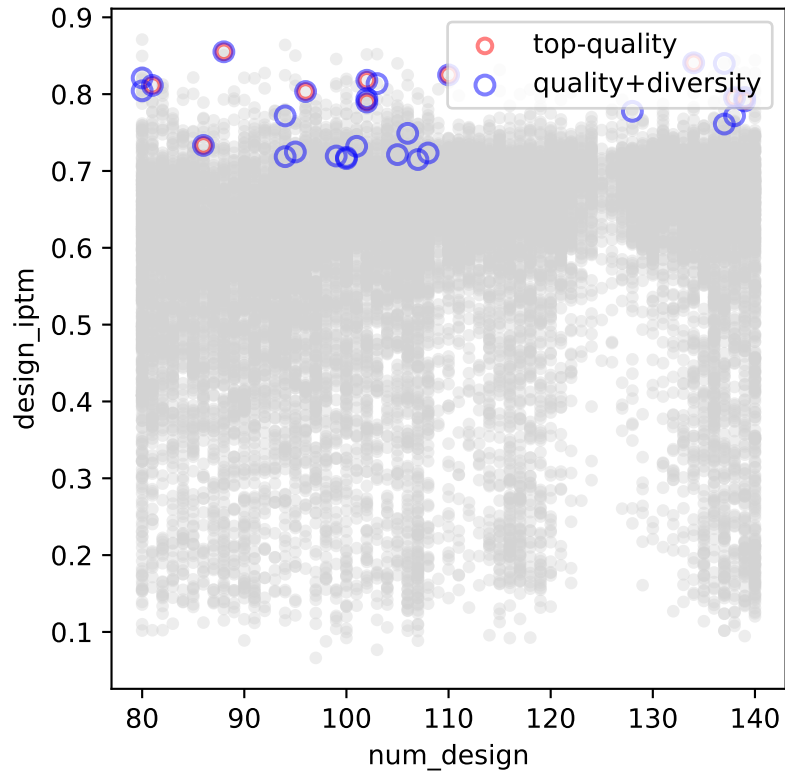
All samples



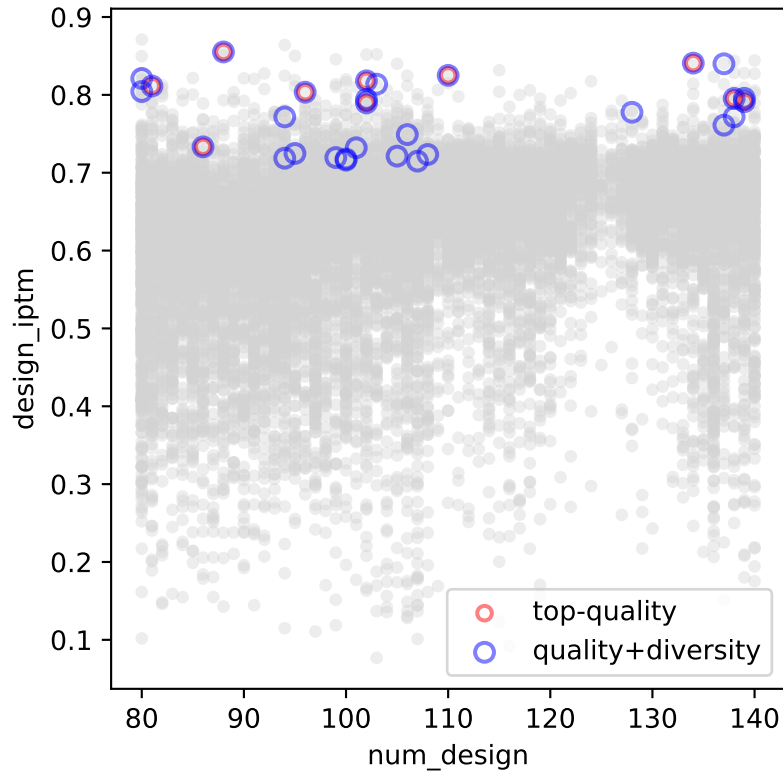
(Designs passing RMSD threshold)



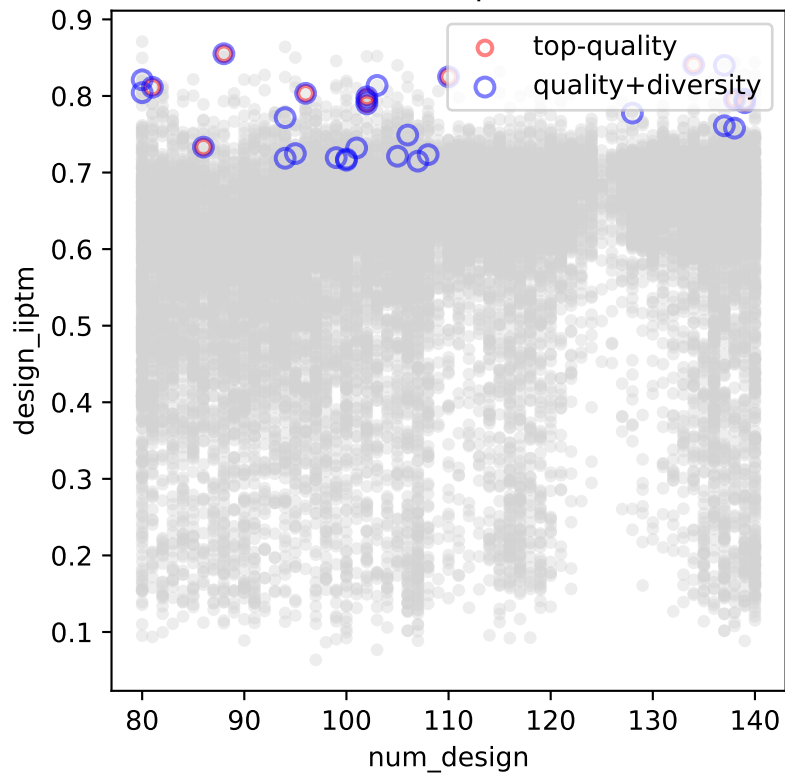
All samples



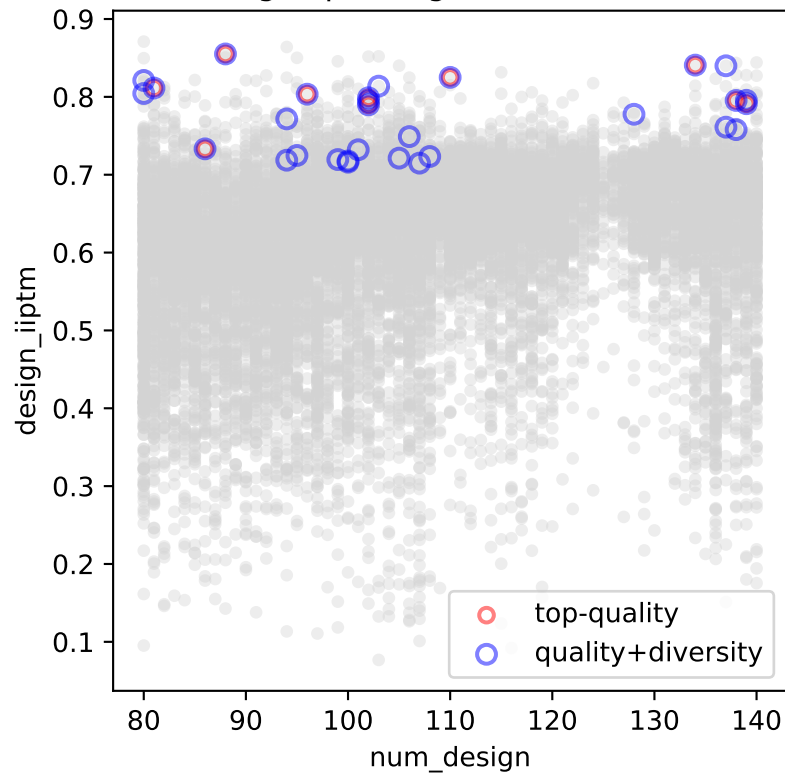
(Designs passing RMSD threshold)



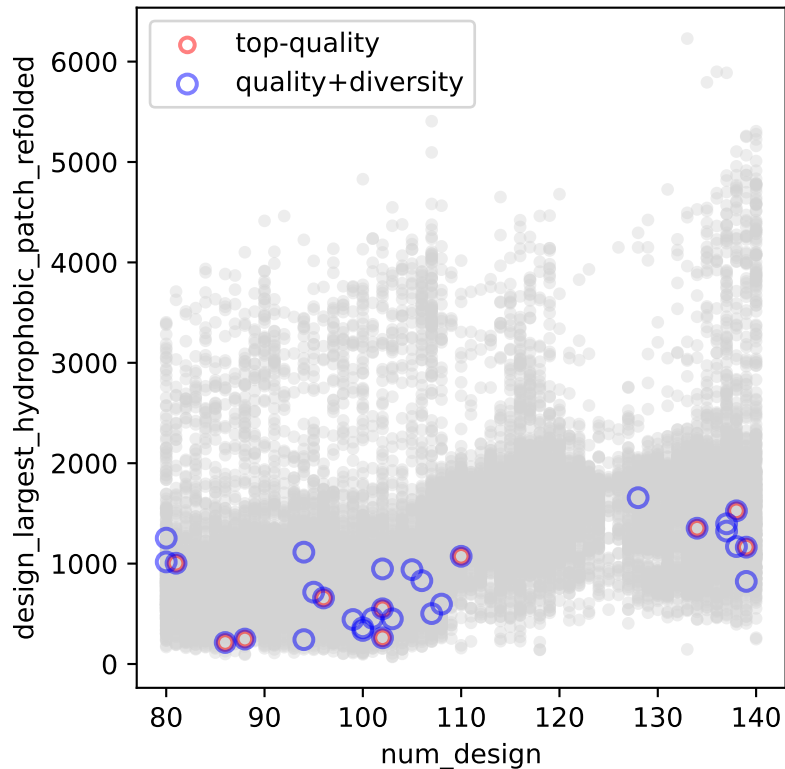
All samples



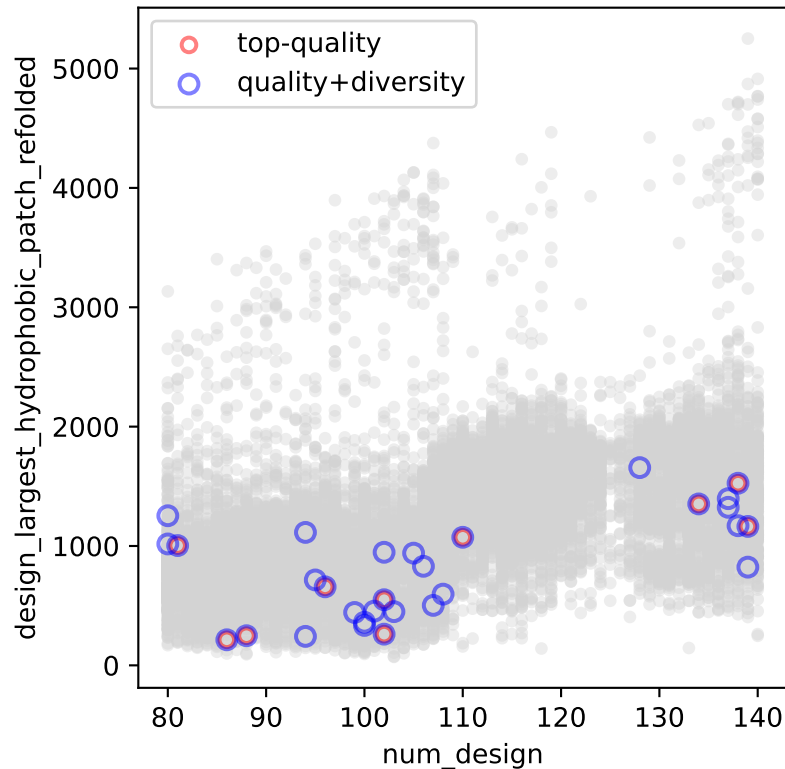
(Designs passing RMSD threshold)



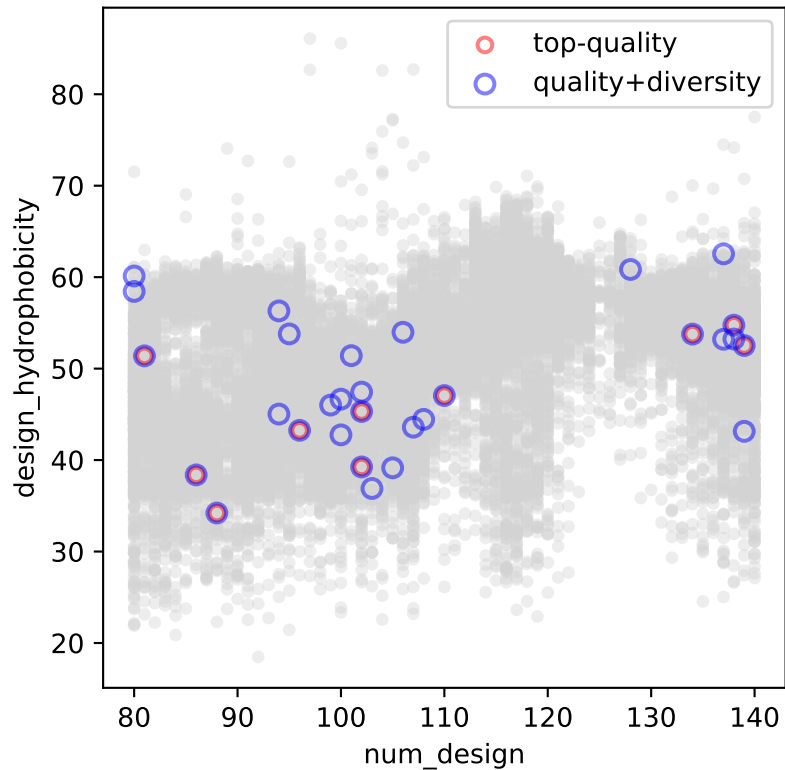
All samples



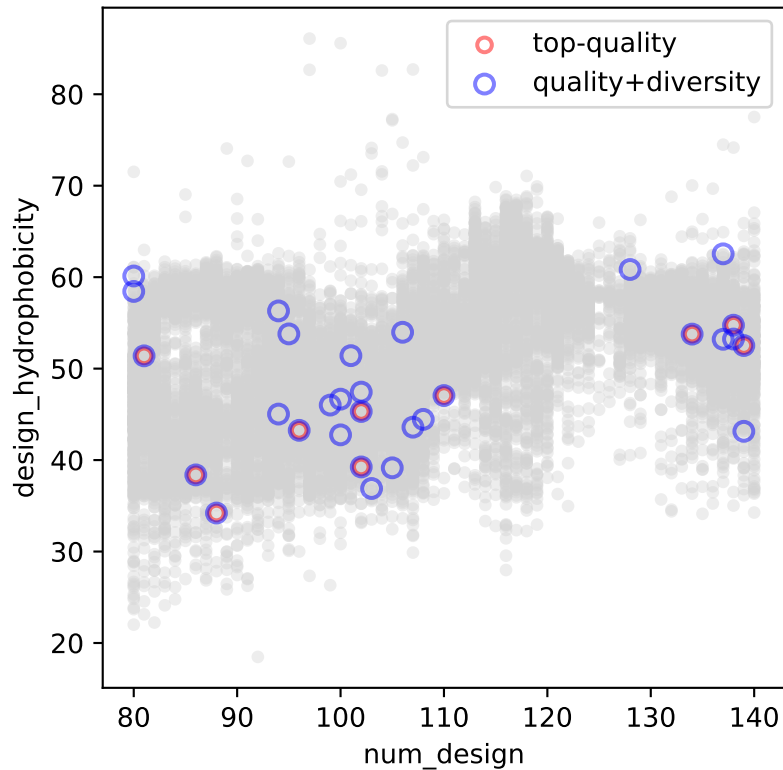
(Designs passing RMSD threshold)



All samples



(Designs passing RMSD threshold)

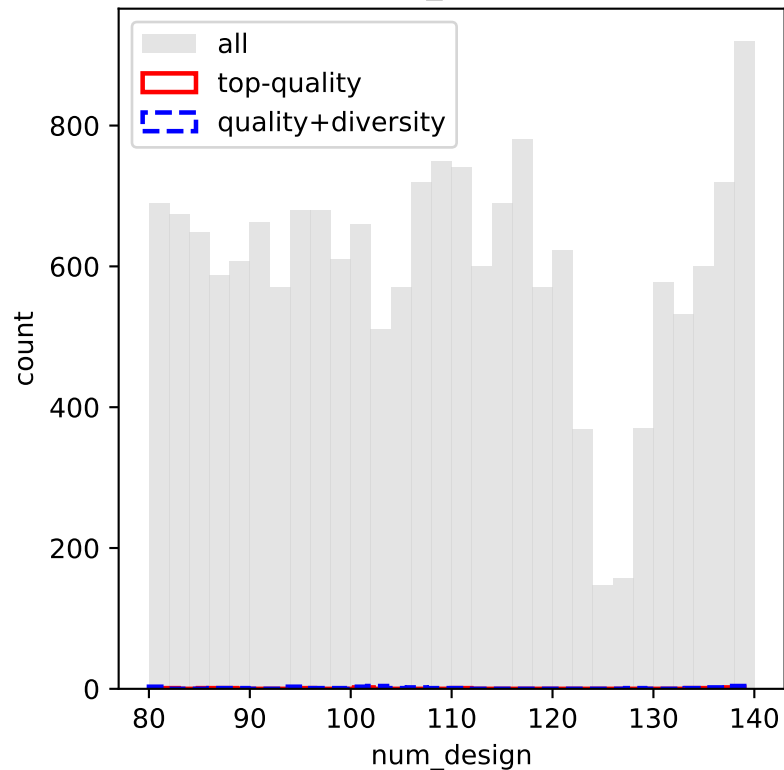


Metric Distributions

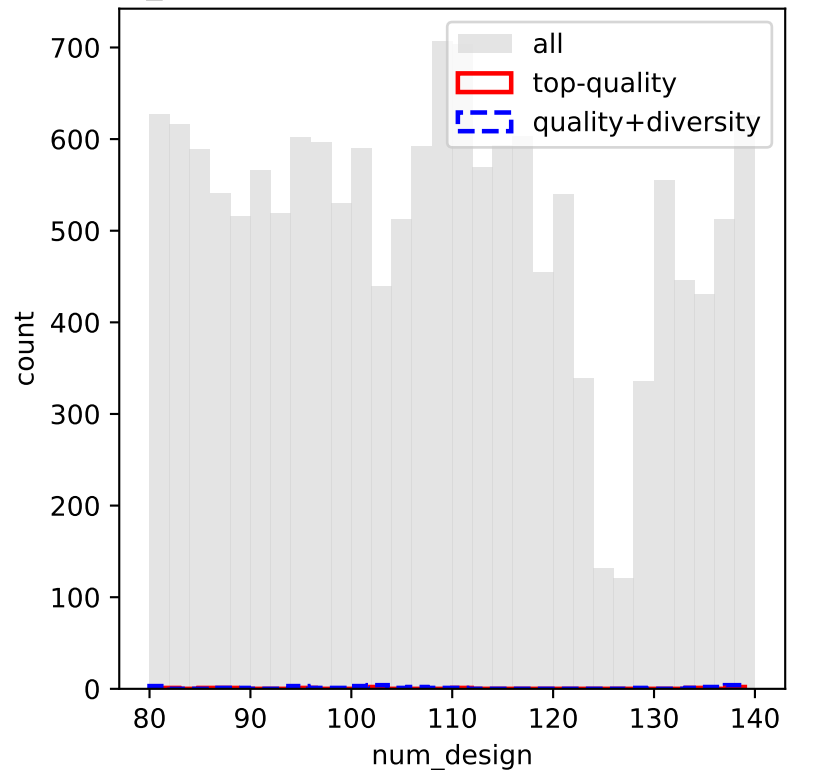
Histograms

Distribution of each metric across all designs (grey) with overlays for Top-quality (red outline) and Diversity-optimised (blue dashed) subsets. The right panel repeats the histogram but only for designs that pass the RMSD threshold.

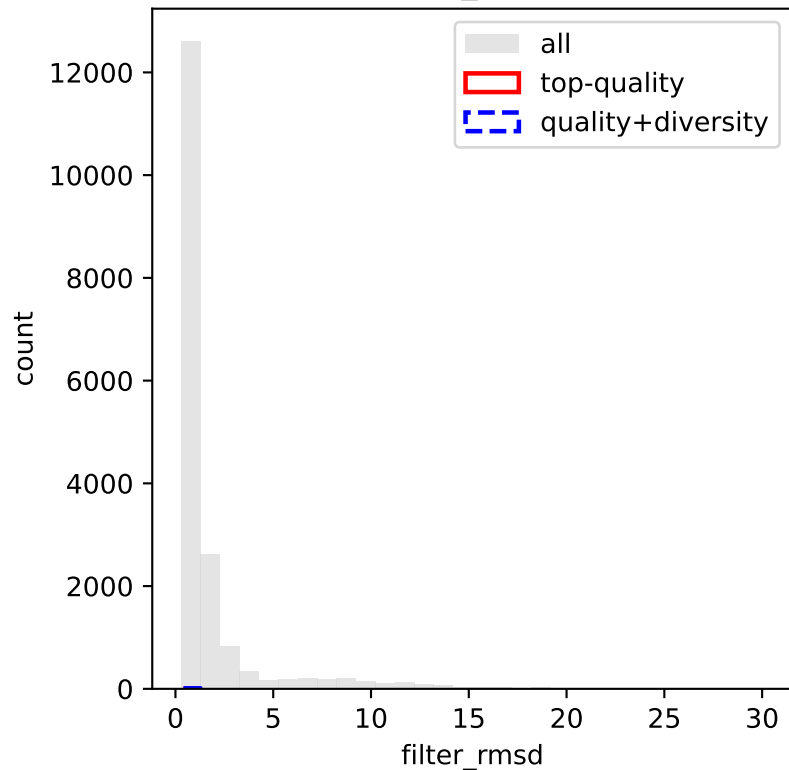
num_design



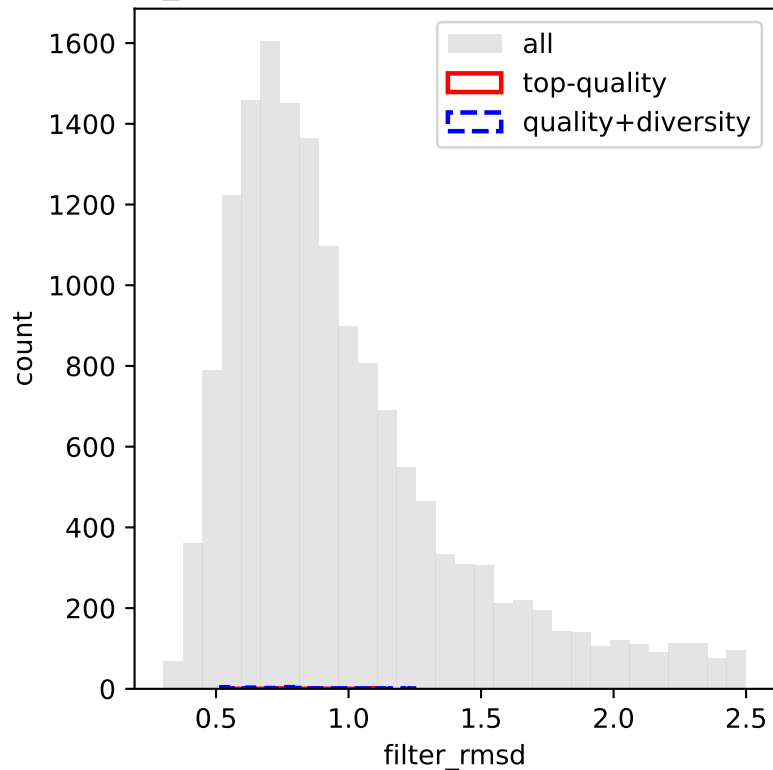
num_design (Designs passing RMSD threshold)



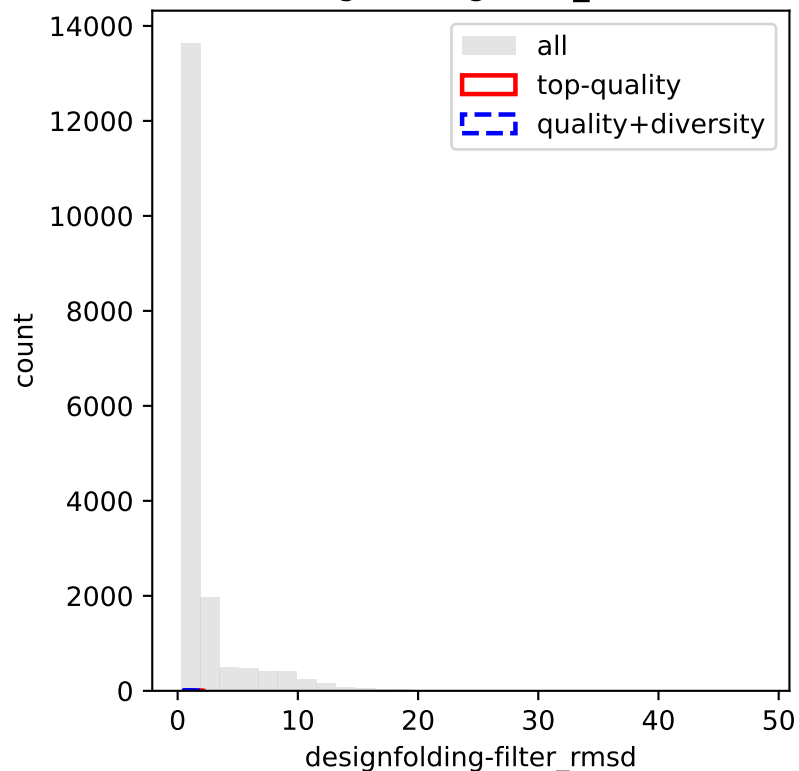
filter_rmsd



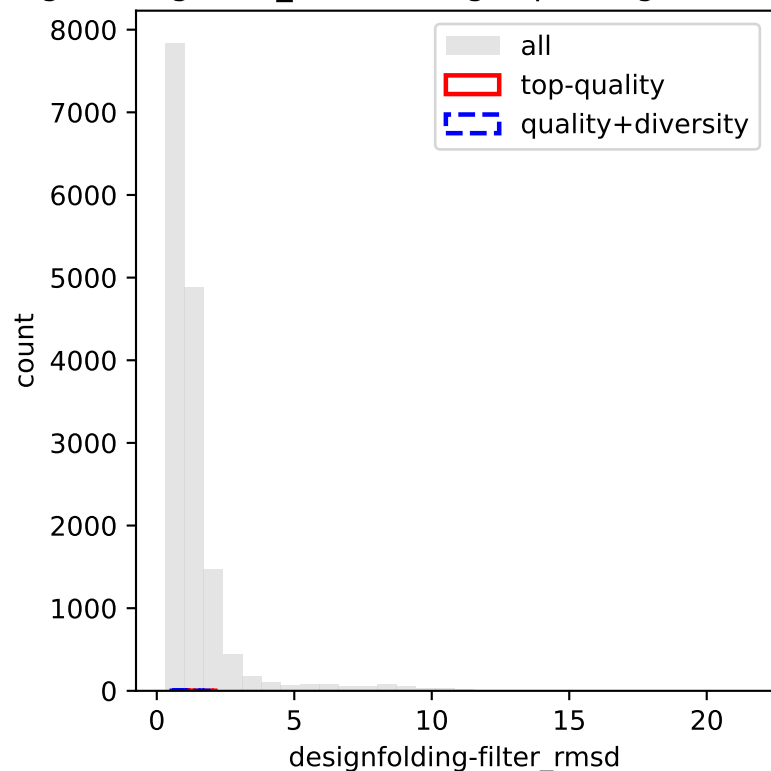
filter_rmsd (Designs passing RMSD threshold)



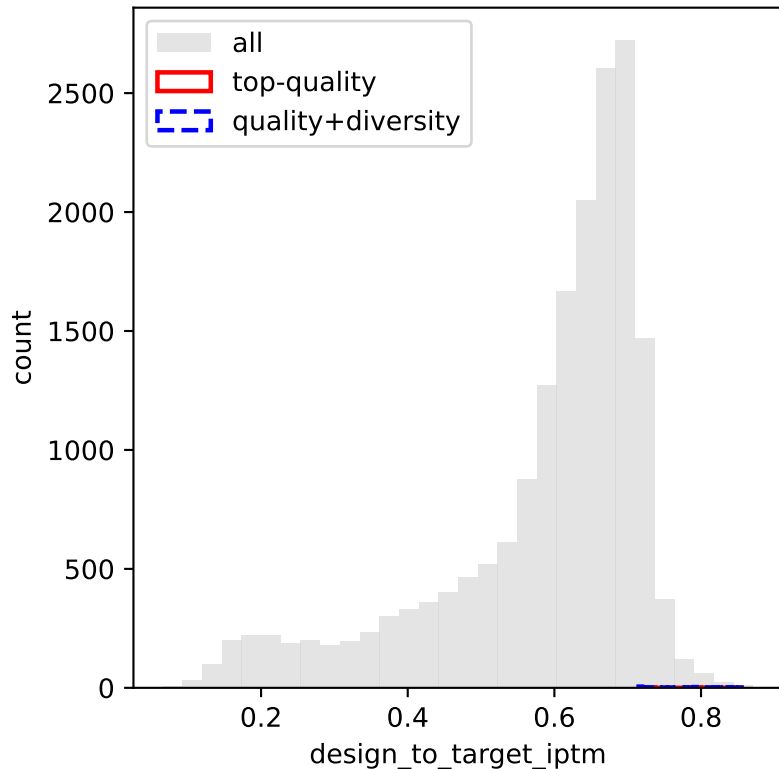
designfolding-filter_rmsd



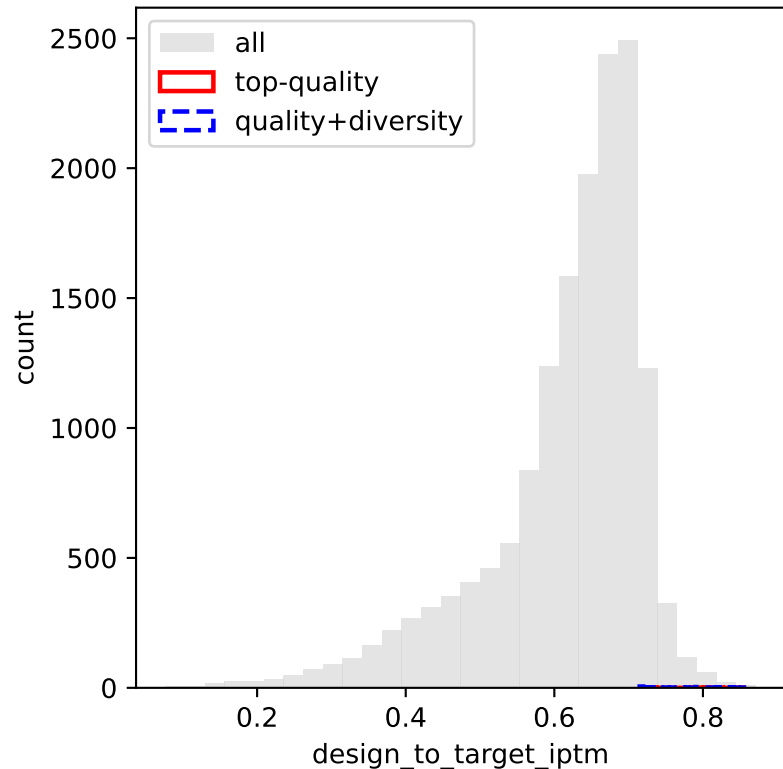
designfolding-filter_rmsd (Designs passing RMSD threshold)



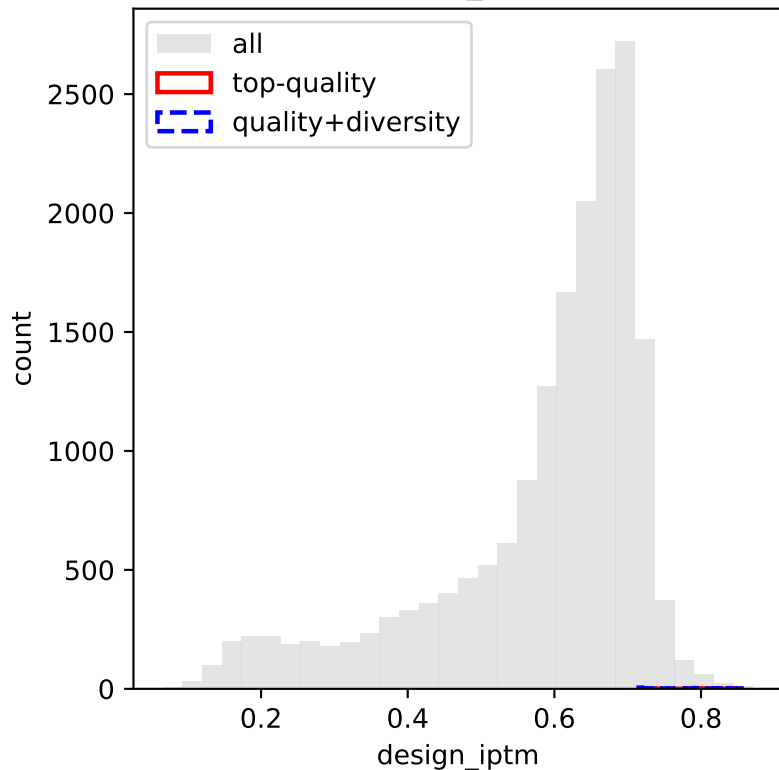
design_to_target_ipbm



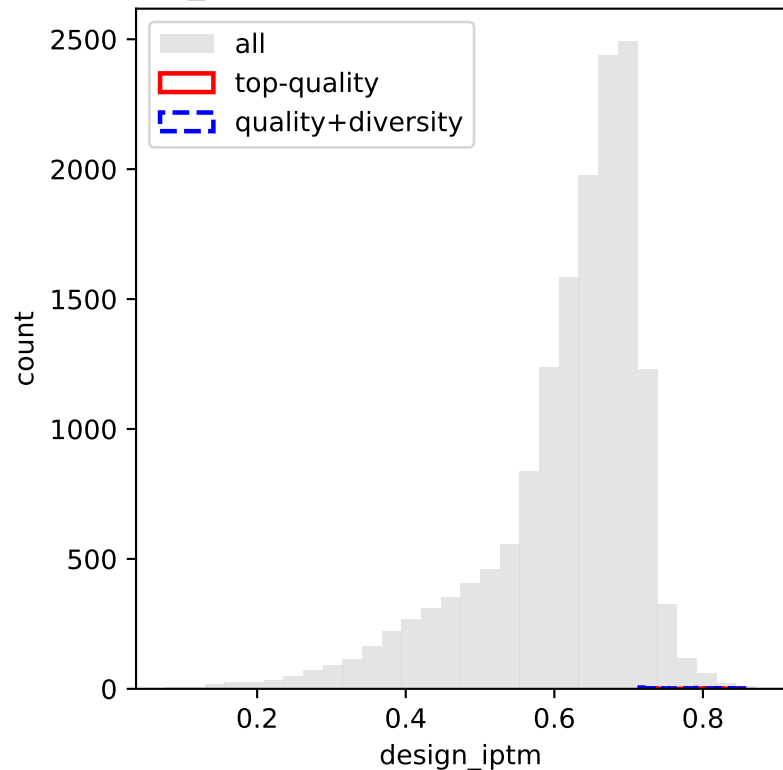
design_to_target_ipbm (Designs passing RMSD threshold)



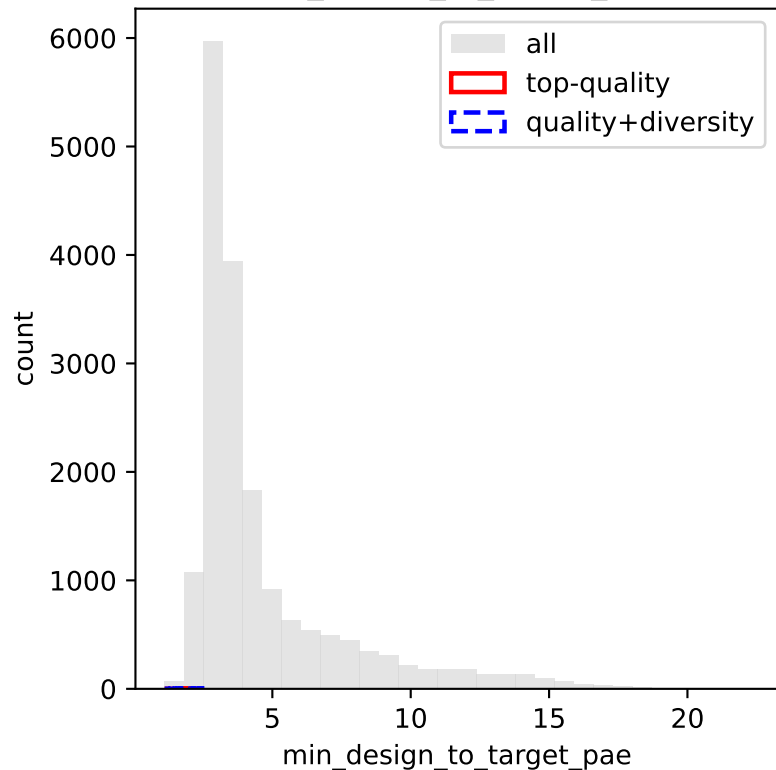
design_iptm



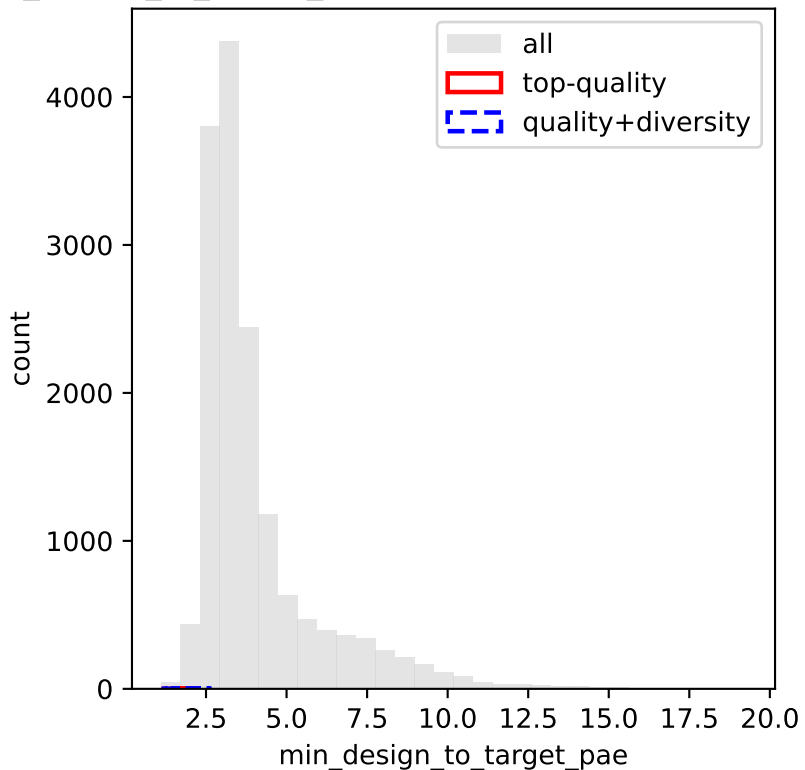
design_iptm (Designs passing RMSD threshold)



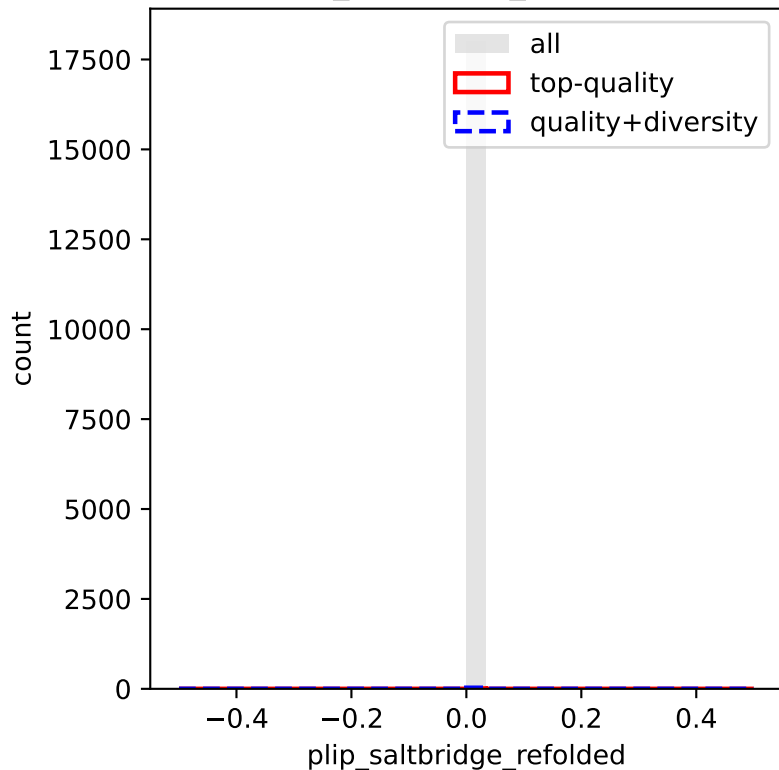
min_design_to_target_pae



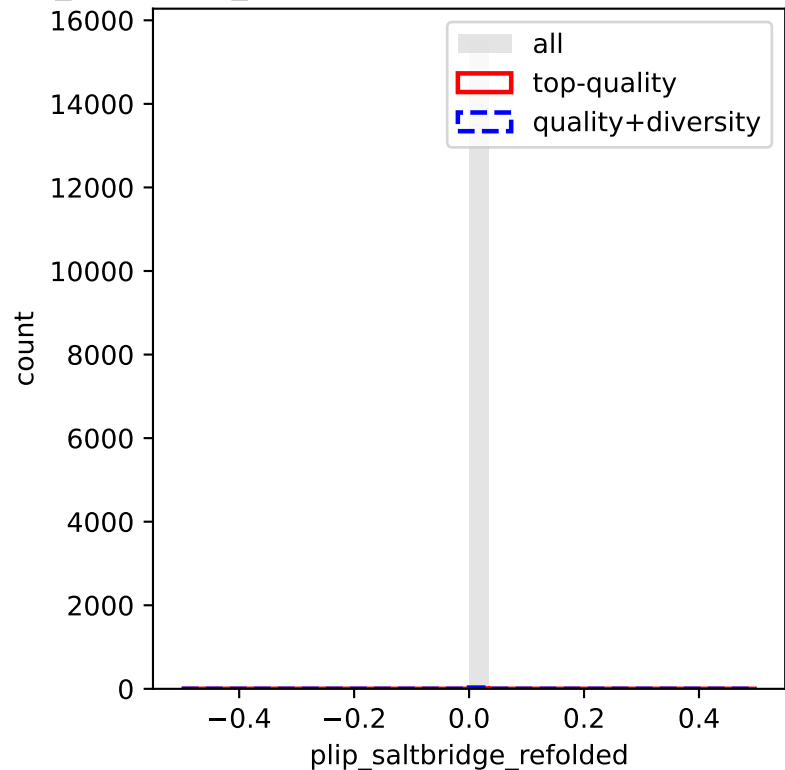
min_design_to_target_pae (Designs passing RMSD th



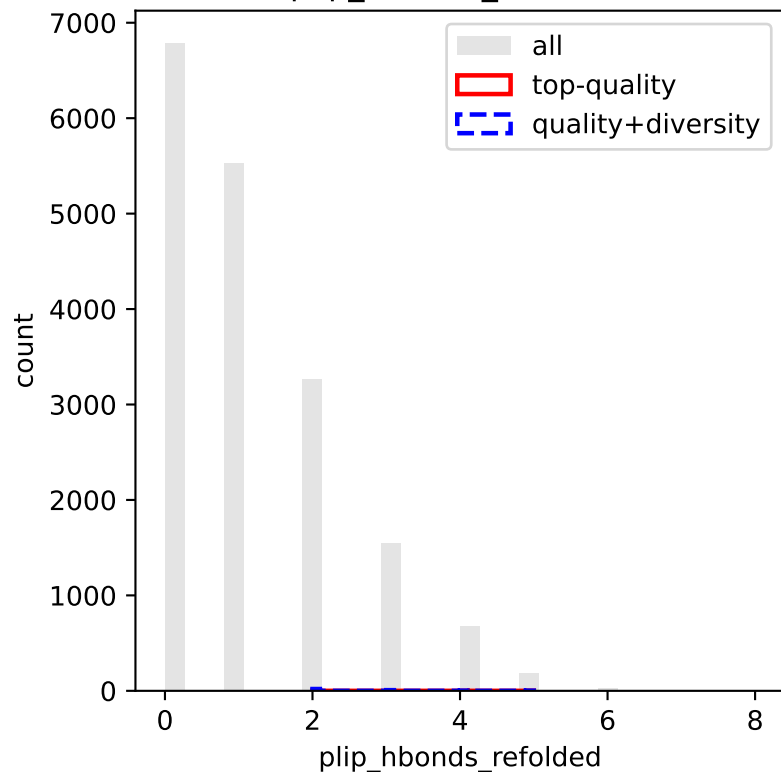
plip_saltbridge_refolded



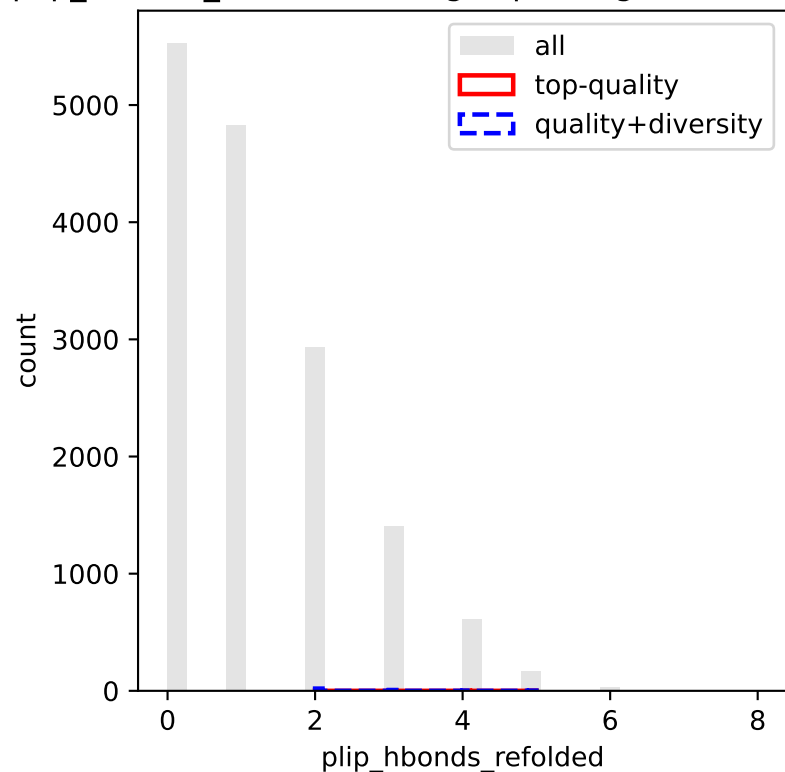
plip_saltbridge_refolded (Designs passing RMSD threshold)



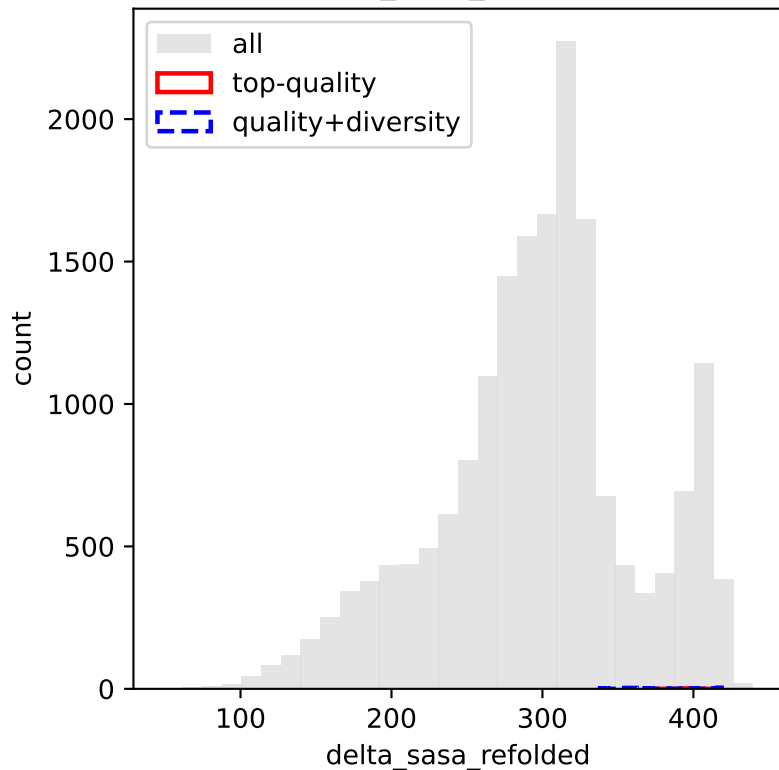
plip_hbonds_refolded



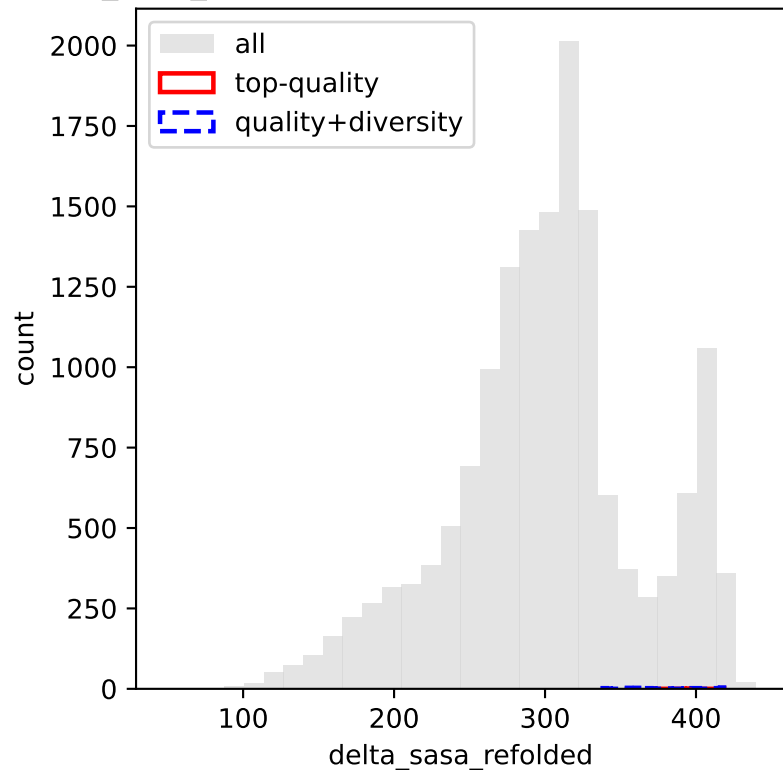
plip_hbonds_refolded (Designs passing RMSD threshold)



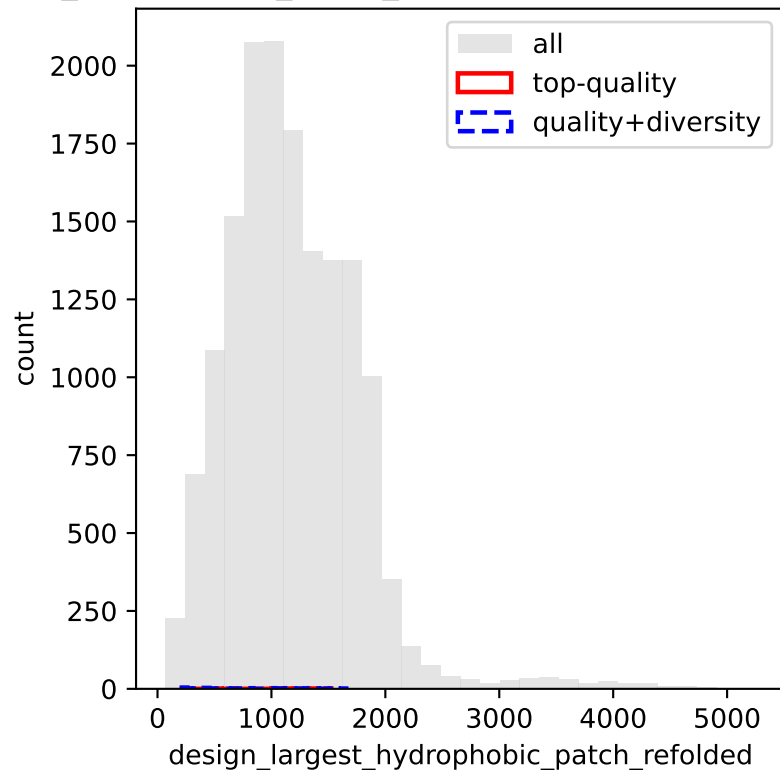
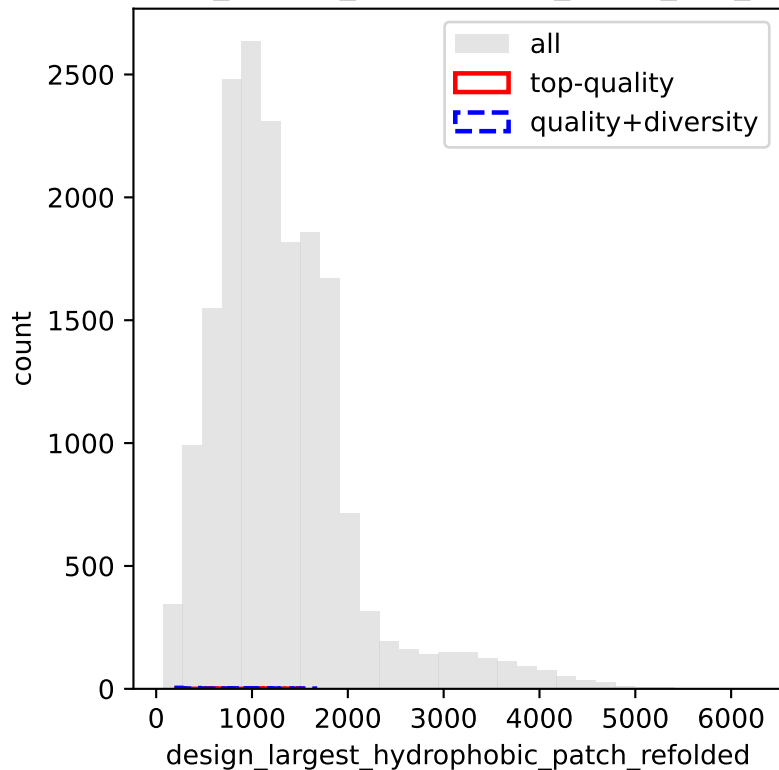
delta_sasa_refolded



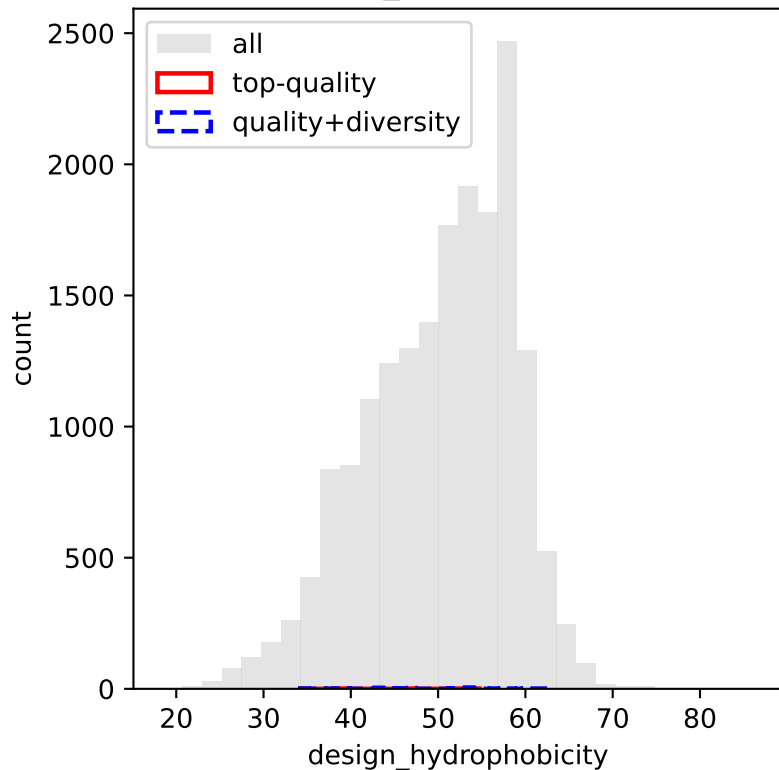
delta_sasa_refolded (Designs passing RMSD threshold)



design_largest_hydrophobic_patch_refolded (Designs passing)



design_hydrophobicity



design_hydrophobicity (Designs passing RMSD threshold)

